

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – INTERACTIVE TREE SIMULATOR

|                  |                                                                                                                                          |               |                                                |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------|
| Course Code:     | CSA0305                                                                                                                                  | Course Title: | Data Structures                                |
| CO Assessed:     | CO3 – Precision (BL3, BL5)                                                                                                               | Assessment:   | Assessment Tool 1 – Interactive Tree Simulator |
| Weightage:       | 30% of CIA                                                                                                                               | Total Marks:  | 100                                            |
| CO3 Statement:   | Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees. |               |                                                |
| Student Name:    | Siva Charan                                                                                                                              | Reg. No.:     | 192525039                                      |
| Section / Batch: | AI and ML/ 2025                                                                                                                          | Date:         | 29/07/2026                                     |

### Code of Conduct

I Siva Charan (192525039) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Siva charan

### Instructions

- This assessment tool contains 5 Interactive Tree Simulator questions, Total: 100 Marks.
- Students can use any online/offline Interactive Tree Simulator. Demonstrate insertion, deletion, searching, and traversal operations wherever applicable.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation.
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

| Section / Topic              | Focus                                                                  | Q Nos     | Marks |
|------------------------------|------------------------------------------------------------------------|-----------|-------|
| Binary Search Tree           | Properties, binary tree and binary search tree, insertion and deletion | 1         | 20    |
| Tree Traversals              | Traversal types, In order, Post order and Pre order                    | 2         | 20    |
| AVL Tree                     | Balancing factor, Rotation, Types of rotation, examples                | 3         | 20    |
| B Tree, 2-3 tree, 2-3-4 tree | Properties, structures and Operations                                  | 4         | 20    |
| Red Black Tree               | Properties, Example and Operations                                     | 5         | 20    |
| GRAND TOTAL:                 |                                                                        | 100 Marks |       |

## Annexure A – Interactive Tree Simulator

- Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80  
Perform:
  - Inorder Traversal
  - Search for 60
  - Delete 30
  - Display the final BST.
- Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80. Find all the traversals.
  - Inorder
  - Preorder
  - Postorder
- Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.
- Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75
- Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree. Analyze their heights.

### Rubrics for Calculation

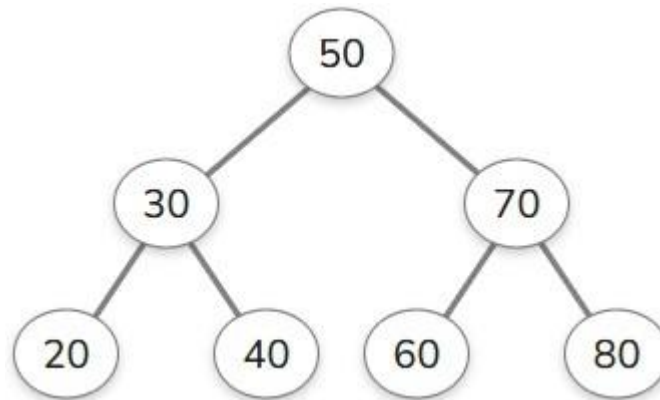
| Criteria                         | Weightage | Level 4 – Exemplary                                                    | Level 3 – Proficient               | Level 2 – Developing                   | Level 1 – Beginning                |
|----------------------------------|-----------|------------------------------------------------------------------------|------------------------------------|----------------------------------------|------------------------------------|
| Correct Tree Construction        | 20        | Tree is constructed correctly with all operations performed accurately | Minor errors in construction       | Some operations incorrect              | Tree construction mostly incorrect |
| Understanding of Tree Operations | 20        | Clearly explains insertion, deletion, search and balancing operations  | Explains most operations correctly | Limited understanding                  | Unable to explain operations       |
| Analysis of Tree Behaviour       | 30        | Correctly analyses balancing, rotations, node split/merge, splaying    | Good analysis with minor omissions | Partial analysis                       | Poor or incorrect analysis         |
| Presentation                     | 20        | Well-organized report with accurate observations and conclusion        | Good report with minor issues      | Basic report with limited observations | Incomplete report                  |
| Accuracy of Responses            | 10        | 90–100% correct answers                                                | 75–89% correct answers             | 50–74% correct answers                 | Below 50% correct answers          |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

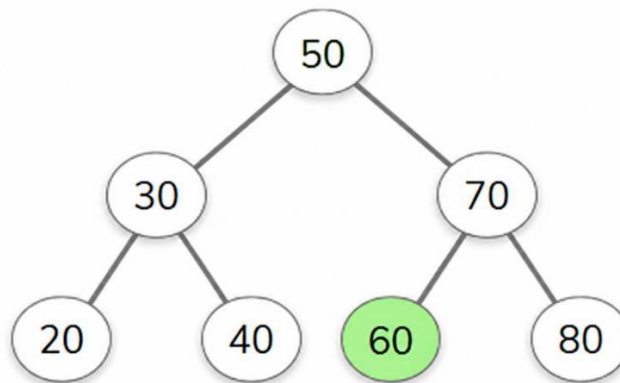
1. Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80

Perform:

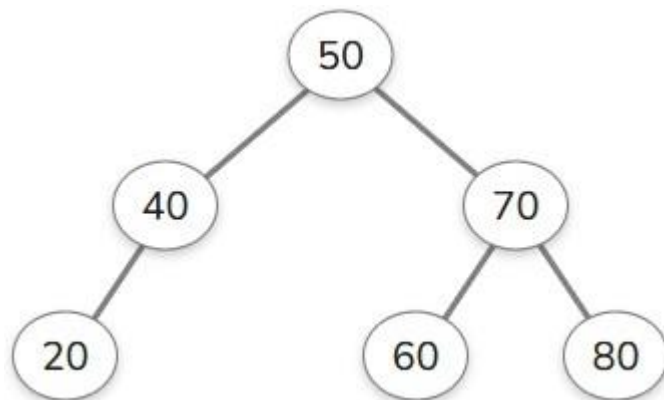
- a. Inorder Traversal



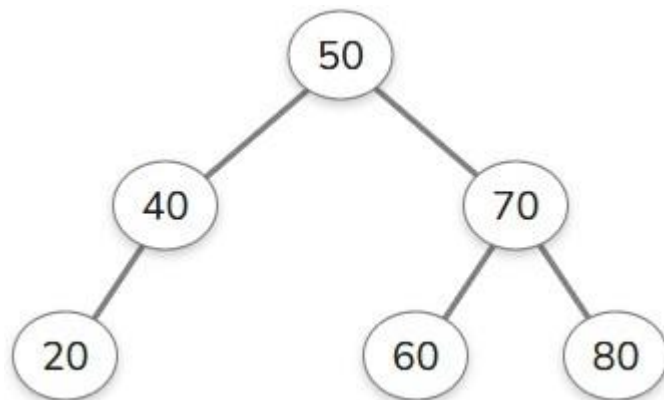
- b. Search for 60



c. Delete 30

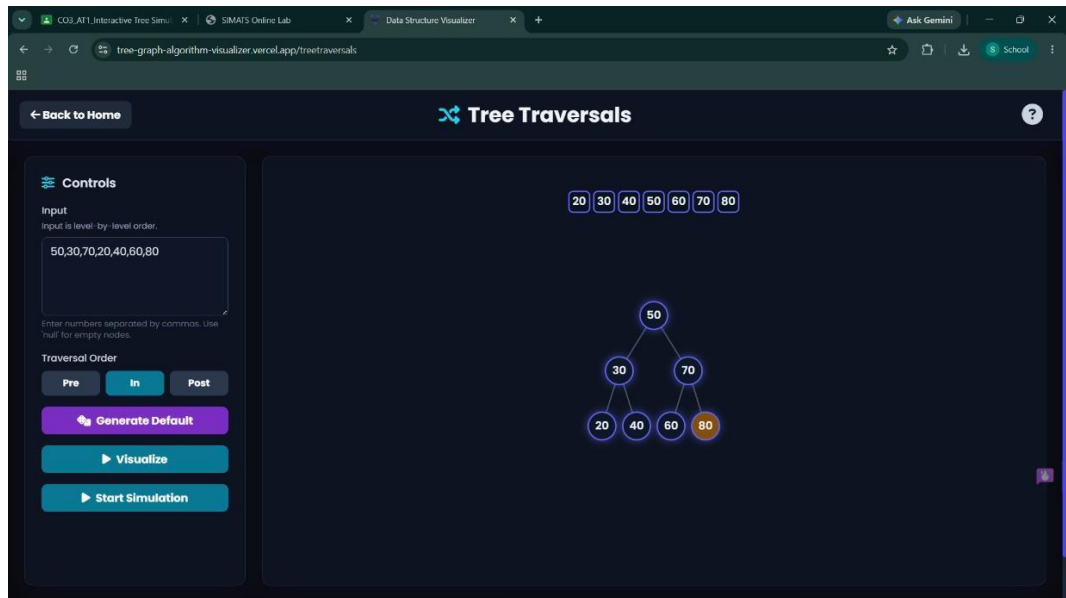


d. Display the final BST.

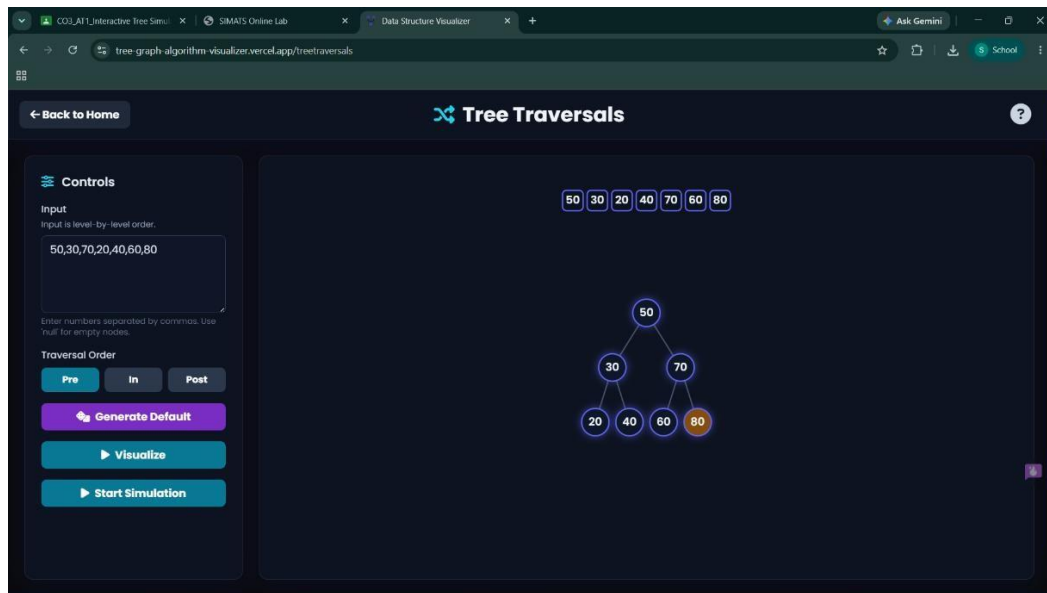


2. Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80. Find all the traversals.

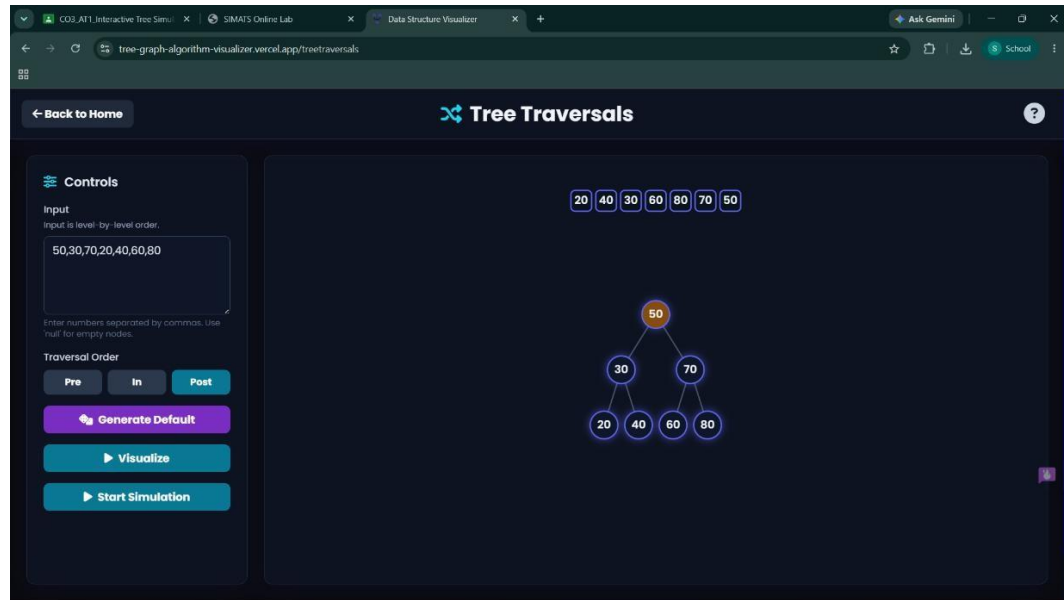
a. Inorder



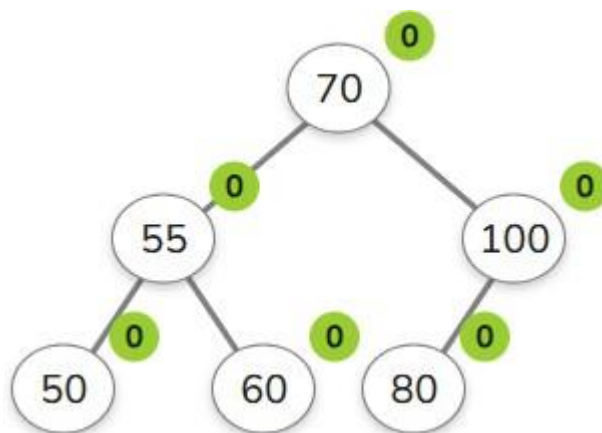
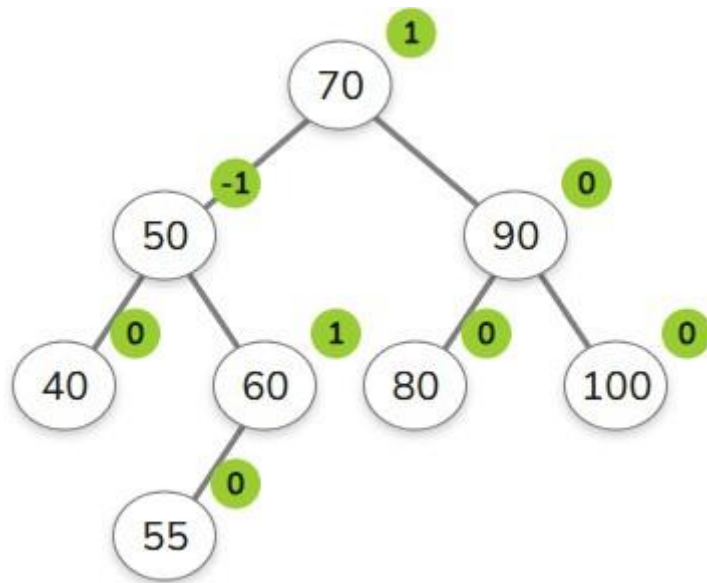
b. Preorder



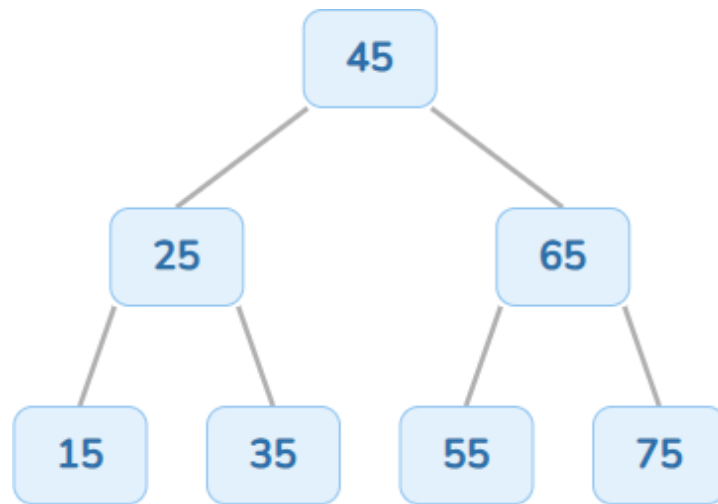
c. Postorder



3. Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.

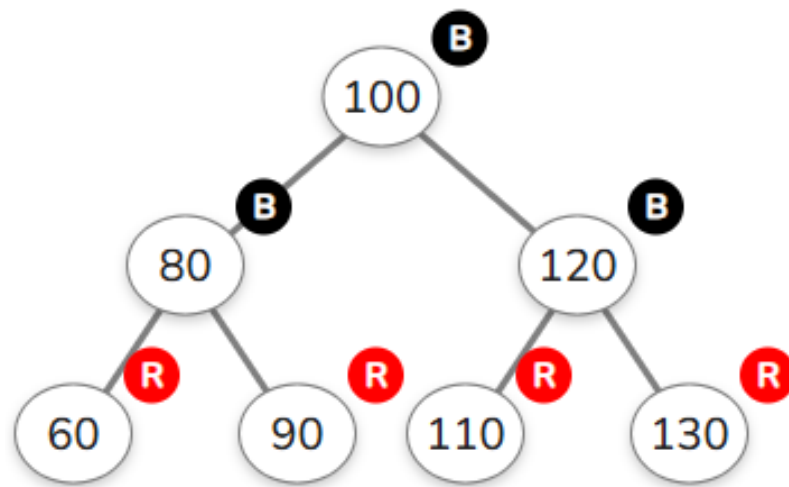


4. Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75





5. Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree. Analyze their heights.



| Insertion | Tree Height |
|-----------|-------------|
| 100       | 0           |
| 80        | 1           |
| 120       | 1           |
| 60        | 2           |
| 90        | 2           |
| 110       | 2           |
| 130       | 2           |